

产 品 规 格 书

SPECIFICATION FOR APPROVAL

客 户 : <https://item.taobao.com/item.htm?id=904918802661&skuid=5932549971534>
(CUSTOMER)

产 品 名 称 : 一体机散热器(G48R)

型 号 : LYP-IR1C4809Y
(MODEL NO.)

規 格 : 9215- Y(0.19A 4000RPM)
(SPECIFICATION)

本公司保证销售之产品均符合本承认书之规格，如有变更则先行送请承认后始得变更
WE GUARANTEE ALL PRODUCTS SOLD ARE UNDER "SAMPLE SPECIFICATION AND APPROVAL SHEET", THERE IS NO CHANGE WILL BE MADE EXCEPT BEEN APPROVED

感谢您给予敝公司送样承认之机会，如惠蒙通过承认，烦请将此表传真回敝公司以利归档
WE APPRECIATE FOR THE OPPORTUNITY OF SAMPLES APPROVAL, PLEASE FAX THIS TABLE BACK TO US FOR FILING AFTER AUTHENTICATED

确 认 APPROVE	审 核 CHECK	制 作 DRAW

客户承认签章 SIGN FOR APPROVED	
日期 DATE	

深圳市利洋鹏科技有限公司

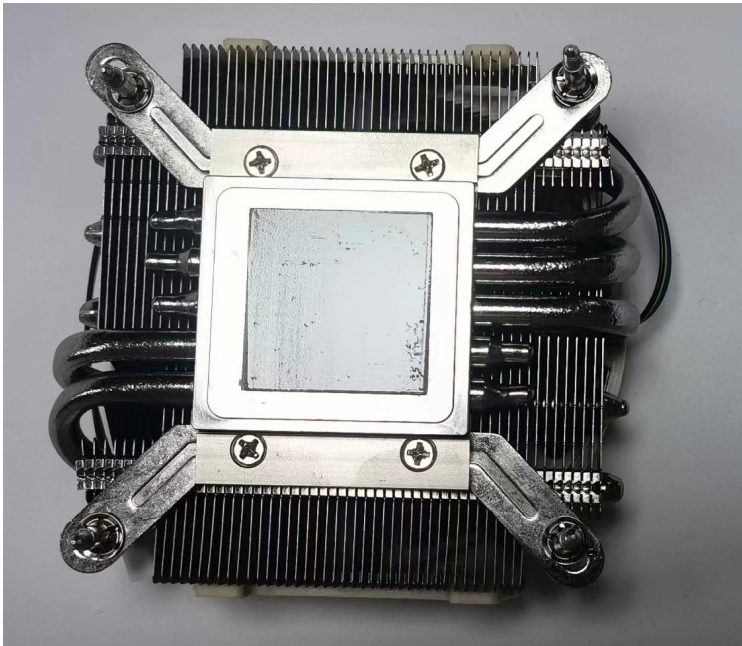
SHENZHEN LI YANG PENG TECHNOLOGY CO., LTD

地址：深圳市龙华区福成街道桔塘社区桔岭新村 269 号祥和园工业区厂房 D 栋 601

ADD : Shenzhen Longhua District Fucheng Street ji Tang Community ji ling New Village 269 Xiangheyuan Industrial Zone Plant D 601

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电邮: hhdin@163.com



利洋鹏科技		G48R 规格书		制作版本: A-1
				页 次: 2 / 19
风 扇 部 分	基 本 参 数	1. 规格: 9215-Y (SPECIFICATION) 2. 外 观 尺 寸: 95*95*15mm (DIMENSIONS) 3. 额 定 电 压: DC12V (RATED VOLTAGE) ±10% 4. 启 动 电 压: 7V (START-UP VOLTAGE) ±10% 5. 额 定 电 流: 0.02-0.19A (RATED CURRENT) ±10% 6. 额 定 功 率: 0.24-2.28 W (POWER CONSUMPTION) ±10% 7. 转 速: 0-4000 P.M. (RATED SPEED) ±10%	8. 最 大 风 量: 14.1-55.26CFM (MAX. AIR FLOW) 9. 最 大 静 压: 0.84-2.87mmH ₂ O (MAX. STATIC PRESSURE) 10. 噪 音: 18.0-44.07dB (A) (NOISE LEVEL) 11. 寿 命: 50,000 hrs (LIFE) 12. 轴承属性: 双滚珠 (TYPE OF BEARING) 13. 极 数: 4 Poles (NO. OF POLE) 14. 回 转 方 向: COUNT-CLOCKWISE (ROTATING DIRECTION)	
	电 气 规 格	风扇旋转锁定测试 (Lock Test)	本公司以 200PCS 测试至少 72 小时, 当锁定解除之后均仍然正常运转。 Locked 200 units for at least 72 hours. Fans work normally after lock released	
		极性保护 Polarity Protect	VCC 与 GND 反接时, 电路不烧毁 (Circuit is protected when VCC GND are exchanged.)	
		绝缘阻抗 (导线) (Insulation Resistance)	10 MΩ /裸线与外框间测量, 500VDC/min (10MΩ /b/w unshackled wire and frame at 500VDC/min)	
		绝缘耐压 (导线) (Dielectric Strength)	5mA Max./导线与外框间测量, 500VAC/min 5mA Max./Measured between lead wire(+)and frame at 500 VAC/min	
	环 境 测 试	运转温/湿度范围 (Operating Temp.Range)	温度/Temperature: -10℃~+75℃ 湿度/ Humidity:35%~85%RH	
		保存温/湿度范围 (Storage Temperature)	温度/Temperature:-20℃~+80℃ 湿度/Humidity:35%~85%RH	
		耐湿性 (Humidity)	试验后, 电气规格依据 MIL-STD 202F Method 103B; 寿命 96 小时; 湿度: 95%RH, 温度: 40±2℃ (Per MIL-STD 202F Method 103B;Life:96Hours; Humidity:95%RH;Temjperature: 40±2℃)	
		包装耐振试验 (Packing Vibration Test)	包装后, 从三方向施 1.1G 加速度, 振动 30 分钟后, 无异状 (Packing Condition X,Y,Z directions 1.1G load vibration for 30min)	
	图	风扇见图纸 (2)		

框
叶

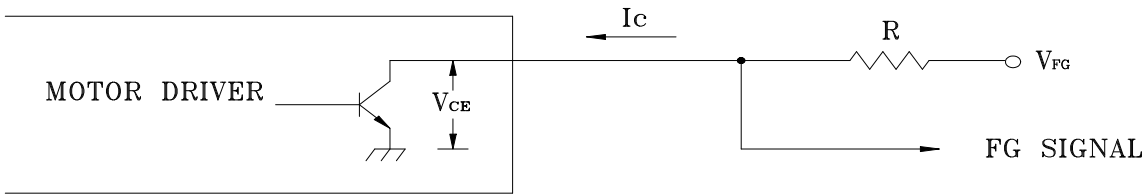
- 1.尺寸---看尺寸图
DIMENSION-- SEE DIMENTION DRAWING
- 2.框架--热塑性的聚酯
FRAME-- THERMOPLASTIC PBT
- 3.扇叶--热塑性的聚酯
FAN BLADE--THERMOPLASTIC PBT

备注：（REMARK）： 见图纸（1）

波
形

FREQUENCY GENERATOR (FG) SIGNAL

1. SCHEMATIC:



CAUTION:

THE LEAD WIRE OF FG SIGNAL CAN NOT TOUCH
THE LEAD WIRE OF POSITIVE OR NEGATIVE.

2. SIGNAL SPECIFICATION:

OUTPUT TYPE: OPEN COLLECT

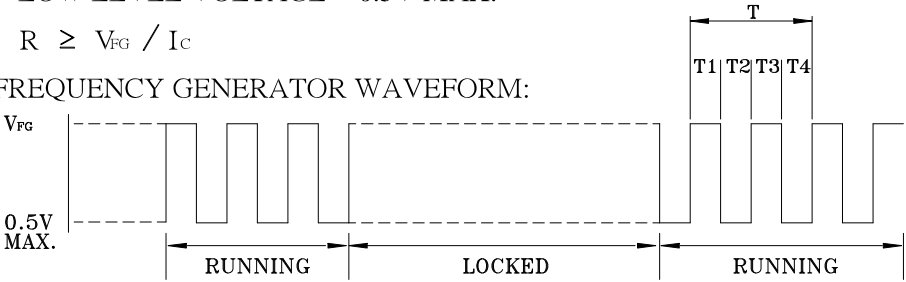
V_{FG} MAXIMUM VOLTAGE = 13.2V

I_c MAXIMUM CURRENT =5mA

LOW LEVEL VOLTAGE = 0.5V MAX.

$R \geq V_{FG} / I_c$

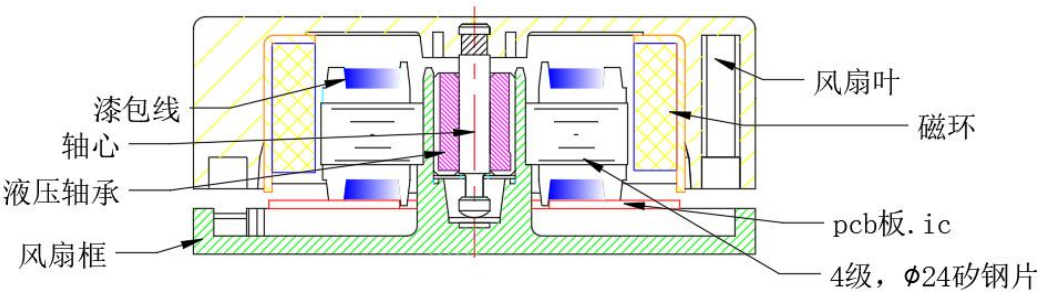
3. FREQUENCY GENERATOR WAVEFORM:

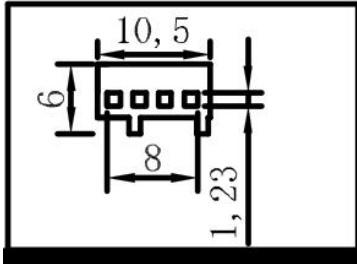
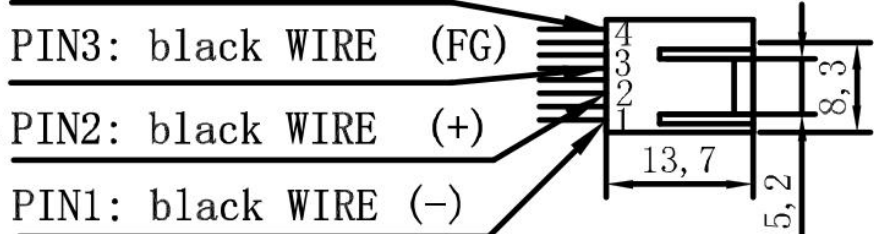
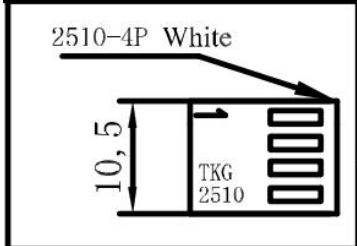


$T=T1+T2+T3+T4=60/N \text{ (Sec)}$

N: SPEED (RPM)

风
扇
结
构
图



利洋鹏科技		G48R 规格书		制作版本：A-1	
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风扇端子接头图面	CONECTOR TYPE:				
			PIN4: black WIRE (PWM) PIN3: black WIRE (FG) PIN2: black WIRE (+) PIN1: black WIRE (-) 		
	2510-4P White 		2510--4pin		
备注：（REMARK）： 导线--: 1007 AWG 26 -: 黑 +:红 FG:黄 PWM:蓝 LEAD WIRE-- 1007 AWG #26 , -: BLACK +: BLACK FG: BLACK PWM: BLACK					
散热片部分	1. 材 质：AL6063T5+CopperC1100 （MATERIAL）				
	2. 铜 重 量：85±5g （WEIGHT）				
	3. 散热片重量：260±5g （WEIGHT）				
	4.材质的化学组成见表（一） （CHEMISTRY COMPOSITION OF MATERIAL）				
	5. 规 格：G48R （SPECIFICATION）				
	6. 制 作 工 艺：铝冲压成型 &铜和热管焊接 （PROCESSOR）				
	7. 颜 色：表面镀镍 （COLOR）				
	8. 外 观 尺 寸：93*102*33.3MM （DIMENSION）				
	9. 图 纸：见图纸（3） （DRAWING）				
配件	1、扣具支架见 (图4)				
	2、螺 杆 . 弹 簧 . 卡 簧 . 螺 丝 见 (图5)				
	3、背板 (图6)				

利洋鹏科技		G48R 规格书		制作版本： A-1
				页 次： 5/19
导热界面	导 热 膏		≧导 热 涂 条	导 热 贴 片
	颜 色： 白色 （Color） 适 用 温 度： -70～+200℃ （Temperature Range） 热 阻： 0.02 （Thermal Resistance）		颜 色： 灰色 （Color） 适 用 温 度： -40～+200℃ （Temperature Range） 热 阻： 0.03 （Thermal Resistance）	颜 色： （Color） 适 用 温 度： （Temperature Range） 热 阻： （Thermal Resistance）
	规格与成分： 见表（二） （SPECIFICATION AND COMPOSITION）			
散热器	1. 重 量： 320±5g （WEIGHT） 2. 外 观 尺 寸： 97*102*48.3mm （DIMENSION） 3. 转 速： 4000±10%RPM （SPEED） 4. 图纸： 见图纸（7） （DRAWING）			
包装部分	内 盒		外 箱	
	1. 材 质： 瓦 楞 （MATERIAL） 2. 外 观 尺 寸： 102*102*50MM （DIMENSIONS） 3. 底 托（Cork base）： 吸塑 （DIMENSIONS）		1. 材 质： 瓦 楞 （MATERIAL） 2. 外 观 尺 寸： 530*530*230MM (45PCS/箱)唛头依客户 A=A （DIMENSIONS）	
散热器整体	成品重量（WEIGHT）： 320±5g 整箱数量（QTY）： 40PCS 净重（N.W）： 13KG 毛重（G.W）： 14KG 整体组装示意图见图纸（7）			

表（一）AL6063T5+CopperC1100 材质的化学组成

合金元素	化学成分（%）			
Cu+Ag	99.91			
P	0.005	合金元素	国家标准（%）	检测值（%）
Fe	0.002	Se	0.2-0.6	0.425
Pb	0.002	Cu	0.1max	0.003
Sb	0.001	Fe	0.35max	0.161
Ag	0.001	Mn	0.10max	0.021
Bi	0.001	Mg	0.45-0.90	0.534
Ni	0.001	Ti	0.10max	＜ 0.10
Zn	0.003	Zn	0.10max	＜ 0.10
Sn	0.001	Cr	0.10max	＜ 0.10
S	0.004	其它单项	0.05max	＜ 0.05
O	0.004	其它总量	0.15max	＜ 0.15
		Al	余量	

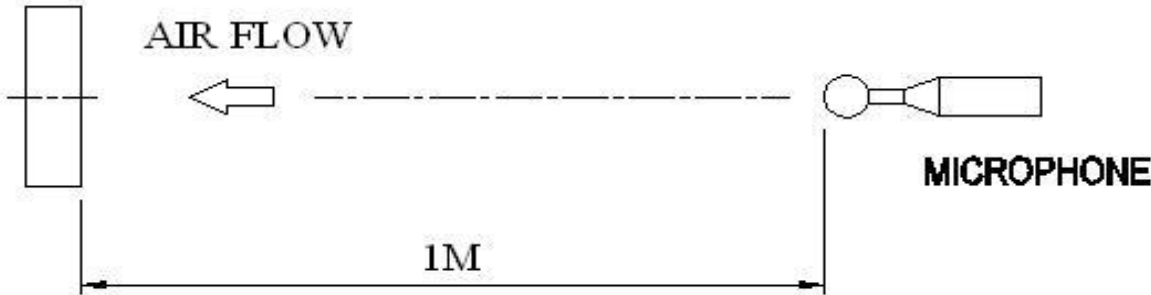
表（二）导热膏 HY810 的规格与成分表

項 目	HY810-C	單位
顏色 Color	GREY	No
熱傳導系數 Thermal Conductivity	>3.0	W/m-K
熱阻抗 Thermal Impedance	<0.058	℃-in² /W
比重 Specific Gravity	>3.15	No
粘度 Viscosity	12500	No
浓度 Thixotropic Index	300±10	1/10mm
瞬 间 耐 温 度 Moment Beared Temperatur	-50~300℃	℃
工作溫度 Operation Temperture	-30~280℃	℃
成 分		
矽化合物 Silicone Compounds	15%	%
碳化合物 Carbon Compounds	35%	%
氧 化 金 屬 化 合 物 Metal Oxide Compoun	50%	%

风扇噪音测试

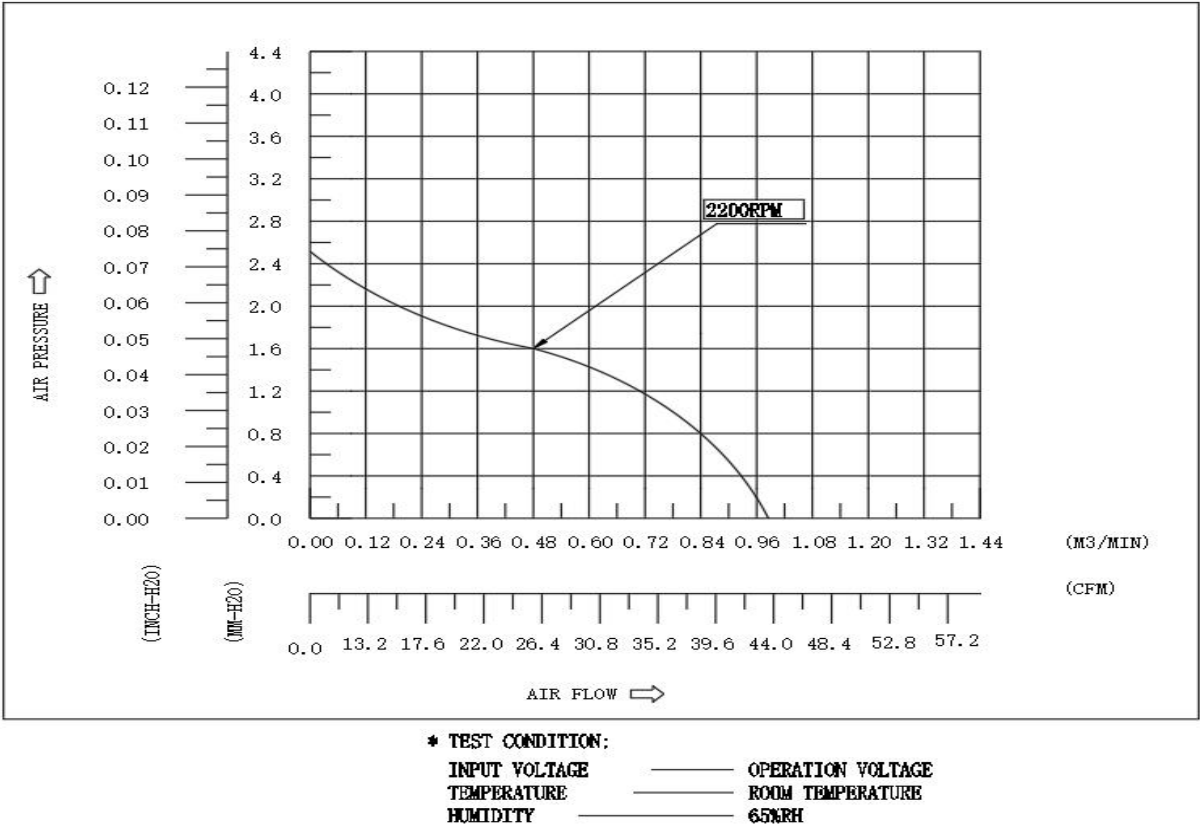
注意:
NOTE:
A. 括号中的数值是有限的参数
THE VALUES WRITTEN IN PARENTHESIS().ARE LIMITED SPEC.
B. 听觉噪音评判条件
ACOUSTICAL NOISE MEASURING CONDITION:

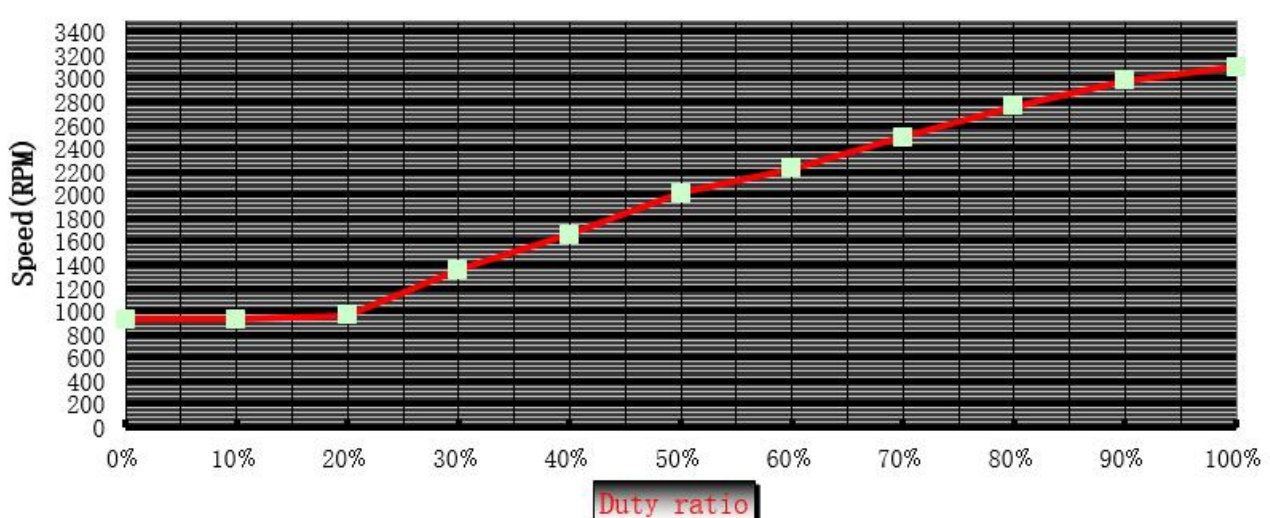
DC FAN



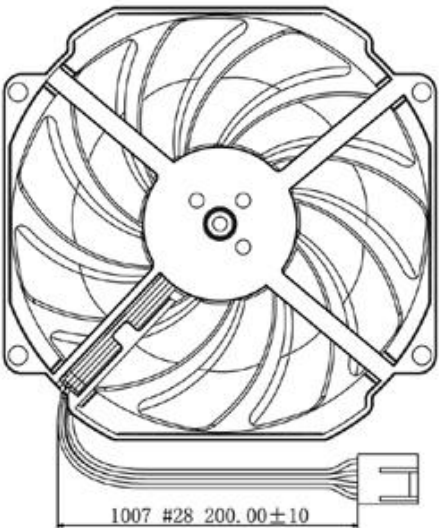
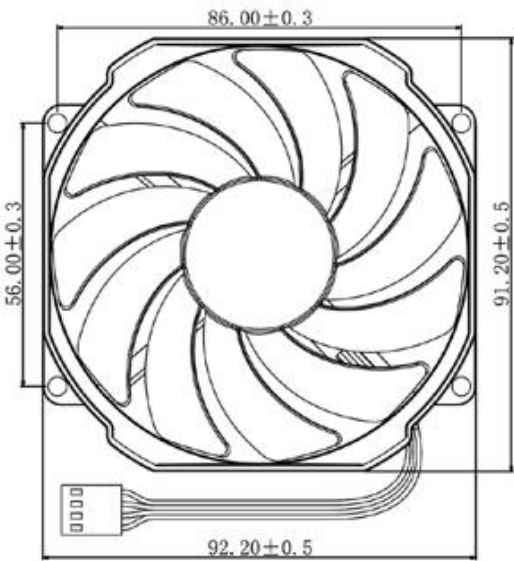
噪声是在额定电压测量在自由空气中进行测量，与拉森戴维斯 824s 类的声级仪一致
NOISE IS MEASURED AT RATED VOLTAGE IN FREE AIR IN ACOUSTICAL CHAMBER WITH LARSON DAVIS TYPE 824S SOUND LEVEL METER.
C. 气流和起亚是在额定电压，参照标准 210-85
THE AIR FLOW AND AIR PRESSURE MEASURED AT RATED VOLTAGE IN DOUBLE CHAMBER IS MEASURED ACCORDING TO AMCA STANDARD 210-85.

风量风压曲线图

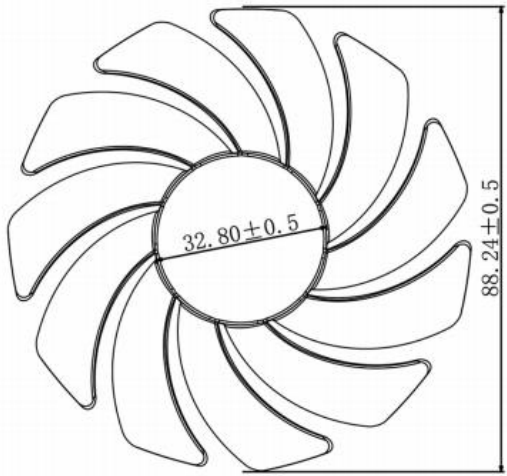
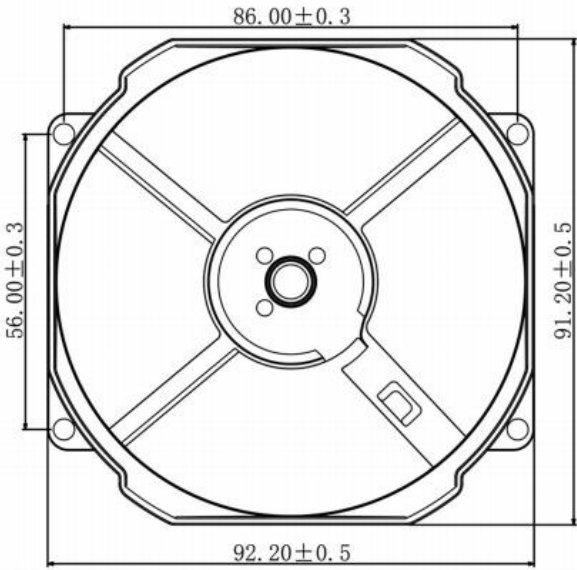


风扇转速曲线图	Accelerated Feature Of PWM Fan Test Report DC风扇转速曲线图 <table><caption>DC Fan Speed vs Duty Ratio Data</caption><tr><th>Duty ratio</th><th>Speed (RPM)</th></tr><tr><td>0%</td><td>900</td></tr><tr><td>10%</td><td>950</td></tr><tr><td>20%</td><td>1000</td></tr><tr><td>30%</td><td>1300</td></tr><tr><td>40%</td><td>1600</td></tr><tr><td>50%</td><td>2000</td></tr><tr><td>60%</td><td>2200</td></tr><tr><td>70%</td><td>2400</td></tr><tr><td>80%</td><td>2600</td></tr><tr><td>90%</td><td>2800</td></tr><tr><td>100%</td><td>3200</td></tr></table>	Duty ratio	Speed (RPM)	0%	900	10%	950	20%	1000	30%	1300	40%	1600	50%	2000	60%	2200	70%	2400	80%	2600	90%	2800	100%	3200
Duty ratio	Speed (RPM)																								
0%	900																								
10%	950																								
20%	1000																								
30%	1300																								
40%	1600																								
50%	2000																								
60%	2200																								
70%	2400																								
80%	2600																								
90%	2800																								
100%	3200																								
PWM 转速测试	PWM DUTY CYCLE VS. RPM <table><tr><th>占空比</th><th>转速</th><th>电流</th></tr><tr><td>0%</td><td>0</td><td>0.02</td></tr><tr><td>50%</td><td>2200</td><td>0.08</td></tr><tr><td>100%</td><td>4000</td><td>0.18</td></tr></table>	占空比	转速	电流	0%	0	0.02	50%	2200	0.08	100%	4000	0.18												
占空比	转速	电流																							
0%	0	0.02																							
50%	2200	0.08																							
100%	4000	0.18																							
PWM 测试条件	1. PWM 频率 = 25 KHZ PWM Frequency PWM 信号振幅 = 5.0 VDC The level of PWM Signal 2. 风扇输入电压 = 12.0 VDC The input voltage																								

图纸
1



图纸 2



单位: mm

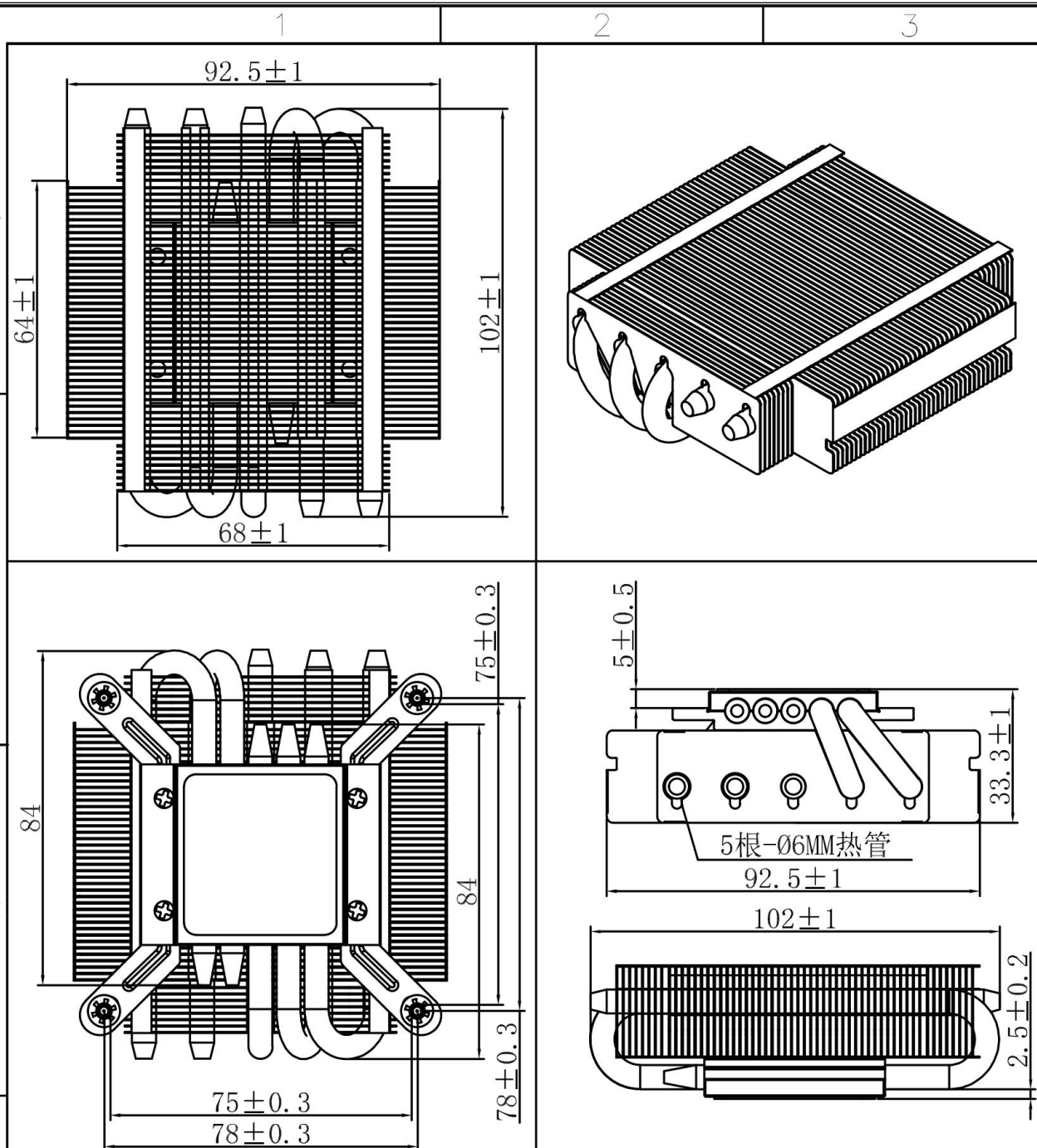
- 注:
- 1. 框为PBT黑色塑胶材料.
 - 2. 无毛边. 无缺料.
 - 3. 轴露7.5mm.
 - 4. 叶为PBT黑色塑胶材料.

	未注公差		批准			产品名称	
		规格	审核			9015风扇框叶	
	0 - 20	±0.1	制图	物料编码			
	20.01 - 50	±0.2					
	50.01 - 80	±0.3	日期	2024.5.25			
		±0.5	比例	单位	颜色	深圳市利洋鹏科技有限公司 SHENZHEN LI YANG PENGTECHNOLOGY CO., LTD	
			1:1	mm	黑色		
	未倒圆角		材质	数量			
		PBT 15%-94V0	pcs				

1

2

3



注：1. 表面镀镍；2. 不能有毛边毛刺，刮伤，变形，阳极不良等现象；

未注公差		批准	
	规格	审核	
0 - 20	± 0.1	制图	D.M.G.H
20.01 - 50	± 0.2	日期	2020.12.13
50.01 - 80	± 0.3	比例	1:1
	± 0.4	单位	mm
未倒圆角		颜色	黑色
		数量	pcs

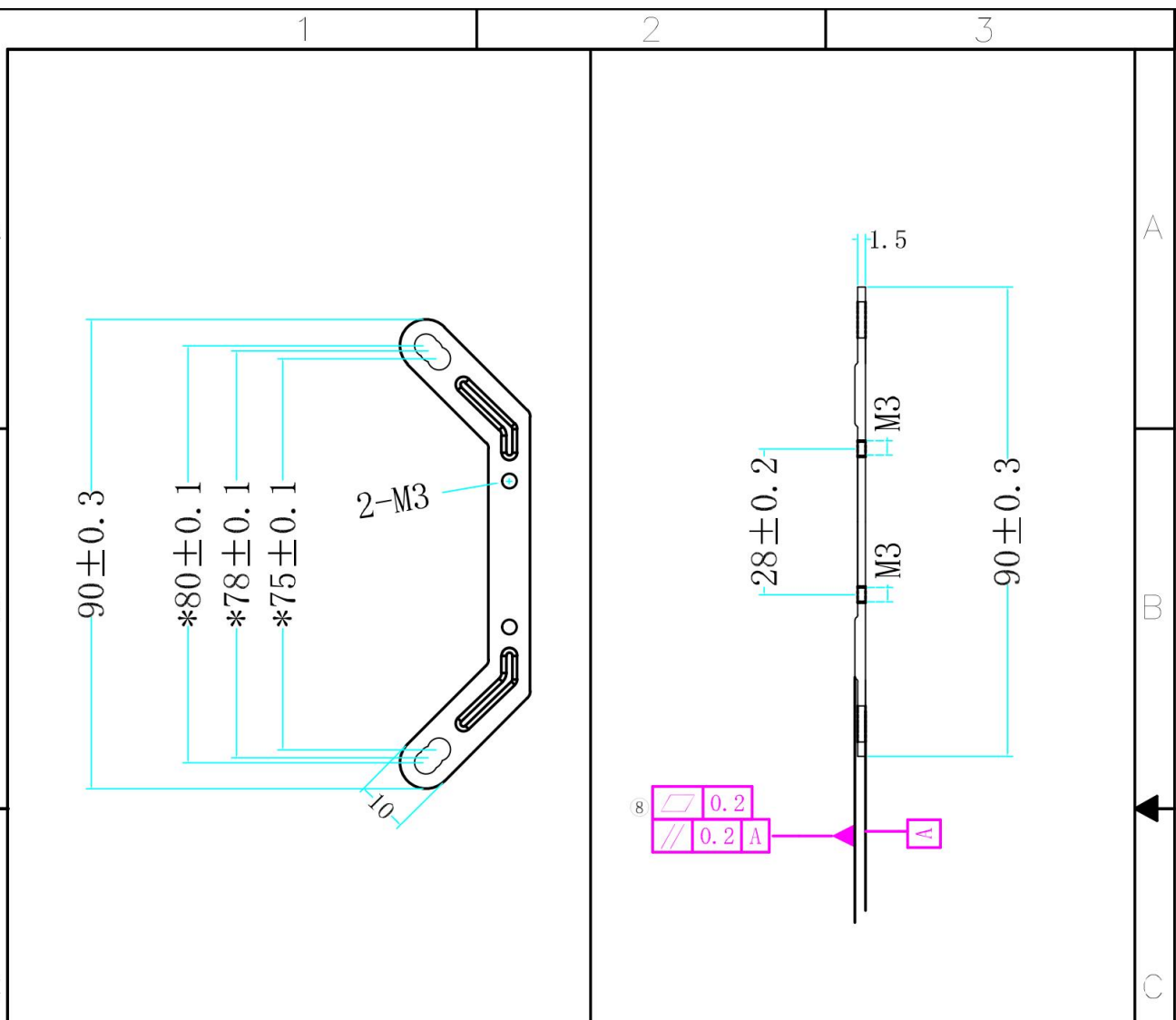
产品名称

G48R散热片

物料编码

G48-5R

深圳市利洋鹏科技有限公司
SHENZHEN LI YANG PENGTECHNOLOGY CO., LTD



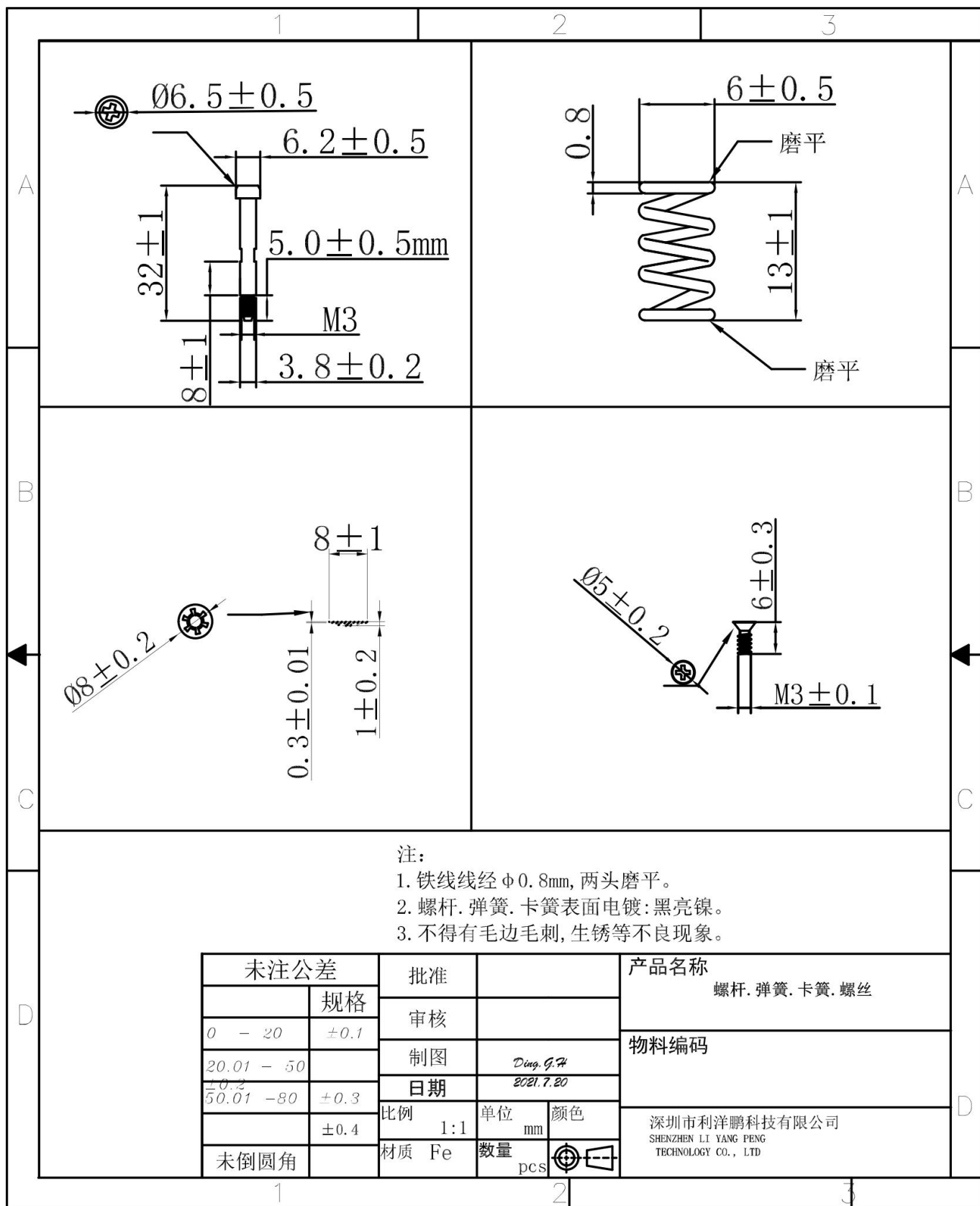
技术要求:

- 1.材质: 碳钢 2.后处理: 电镀 3.硬度: 180 ± 20 Hv
4.板厚: 1.5 ± 0.5 mm 5.除去毛边及锋边

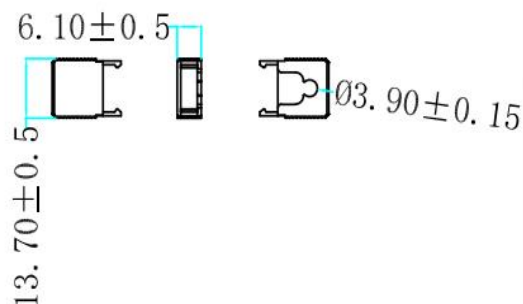
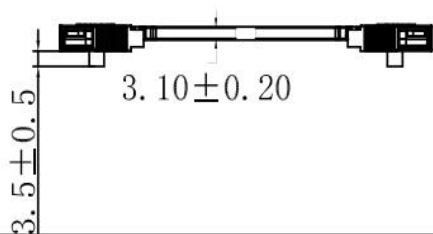
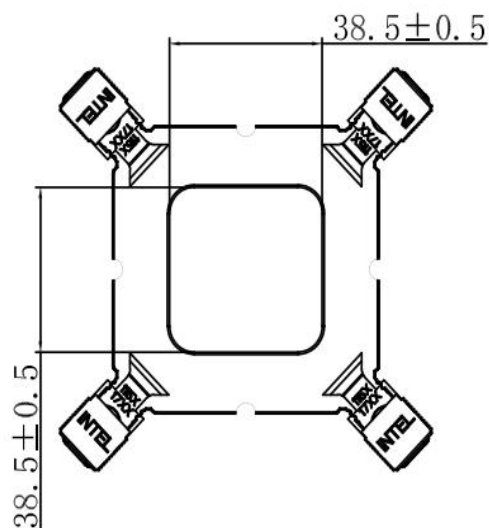
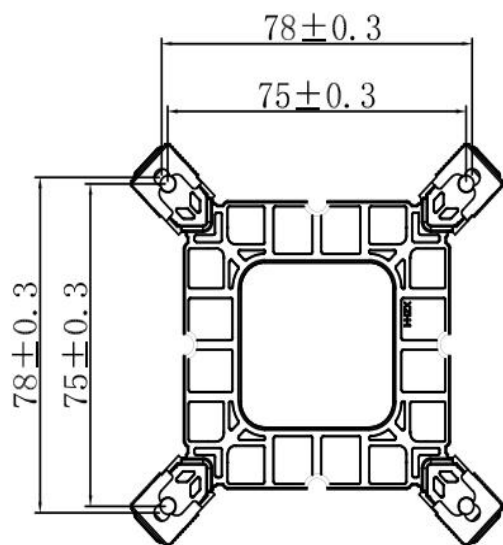
未注公差		批准	产品名称	
	规格	审核	扣具	
0 - 20	± 0.1	制图	物料编码	
20.01 - 50		日期	2021.7.20	
± 0.2		比例	单位	颜色
50.01 - 80	± 0.3	1:1	mm	
	± 0.4	材质	数量	
未倒圆角		Fe	pcs	

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图纸 6



注:

1. 螺母表面电镀: 银亮镍。
2. 塑胶无毛边, 毛刺。

未注公差		批准	产品名称	
	规格	审核	多用背板	
0 - 20	±0.1	制图	物料编码	
20.01 - 50		日期		
±0.2		比例	深圳市利洋鹏科技有限公司 SHENZHEN LI YANG PENG TECHNOLOGY CO., LTD	
50.01 - 80	±0.3	1:1		
	±0.4	材质 Fe	数量 pcs	
未倒圆角		PBT 15% 94V0		



Verification of Conformity

Certificate No. : HSO170109040NCF
Applicant : Shenzhen Liyangpeng Technology Co.,Ltd.
Address : Floor 6, Buliding, No.2 Lingzai, Juling New Village, Jutang Community, Guanlan Street, Longhua New District, Shenzhen
Manufacturer : Shenzhen Liyangpeng Technology Co.,Ltd.
Address : Floor 6, Buliding, No.2 Lingzai, Juling New Village, Jutang Community, Guanlan Street, Longhua New District, Shenzhen
Product : DC FAN
Brand Name : N/A
Model(s) : 9225,3510,4010,4020,4510,4520,5015,5020,5025,6025,6028,7015,7025,7528,7530,8010,8015,8020,8025,8028,8038,7010,7015,7020,7025,7530,8010,8015,8020,8025,8032,8038,9215,9015,9020,9025,9232,9238,9733,12020,12025,12032,12038,14025,15051,17251,20020,20060,12032
Test Report NO. : HSO170109040NRF

Applicable Standard

FCC Part 15:2015 Subpart B

The EUT has been tested and verified to comply with FCC Part 15 Subpart B, Class B digital device requirements.

The EUT has tested to conform to the applicable FCC Rules and Regulations. The method of testing was in accordance to the most accurate measurement standards possible, and that all necessary steps have been enforced to assure that all production units of the same equipment will continue to comply with the Federal Communications Commission's requirements.



This certificate of conformity is based on a single evaluation of the submitted sample(s) of the above mentioned product. It does not imply an assessment of the whole production and other relevant directives have to be observed.

Shenzhen HSO. Test Technology Co., Ltd.

4F, Building 6, Nanxiong Industrial Zone Block 1, Fuhai Avenue East, Fuyang Street, Bao'an District, Shenzhen, China
Web: <http://www.hso-test.com> Tel: +86-755-29484580 Fax: +86-755-29484580



CERTIFICATE OF CONFORMITY

Certificate No. : HSO170109038LCE
Applicant : Shenzhen Liyangpeng Technology Co.,Ltd.
Address : Floor 6, Buliding, No.2 Lingzai, Juling New Village, Jutang Community, Guanlan Street, Longhua New District, Shenzhen
Manufacturer : Shenzhen Liyangpeng Technology Co.,Ltd.
Address : Floor 6, Buliding, No.2 Lingzai, Juling New Village, Jutang Community, Guanlan Street, Longhua New District, Shenzhen
Product : DC FAN
Brand Name : N/A
Model(s) : 9225,3510,4010,4020,4510,4520,5015,5020,5025,6025,6028,7015,7025,7528,7530,6010,6015,6020,6025,6028,6038,7010,7015,7020,7025,7530,8010,8015,8020,8025,8032,8038,9215,9015,9020,9025,9232,9238,9733,12020,12025,12032,12038,14025,15051,17251,20020,20060,12032
Test Report No. : HSO170109038LRE
Test standards : EN 55022:2010+AC:2011
EN 55024:2010+A1:2015

The EUT described above has been tested by us with the listed standards and found in compliance with the council EMC Directive 2014/30/EU. It is possible to use CE marking to demonstrate the compliance with this EMC Directive.

CE



For Chief Executive Frank Liu
Date: Jan.16,2017

This certificate of conformity is based on a single evaluation of the submitted sample(s) of the above mentioned product. It does not imply an assessment of the whole production and other relevant directives have to be observed.

Shenzhen HSO. Test Technology Co., Ltd.
 Rm 425-428,4F,Jimeihu Building, No. 339,107 Country Road, Xixiang Street, Bao'an District, Shenzhen City, P.R.C
 Web: <http://www.hso-test.com> Tel: +86-755-29484580 Fax: +86-755-29484580

CPST

CERTIFICATE OF CONFORMITY

NO.: C211016063001-2

The following products have been tested by us and found in conformity with the RoHS Directive 2011/65/EU Annex II amending Annex (EU)2015/863 It is possible to use CE marking to demonstrate the conformity with this Directive.

Applicant : Shenzhen Liyangpeng Technology Co., Ltd

Address : 601, Zone Plant D, 269 Xiangheyuan Industrial, ji ling New Village, ji Tang Community, Fucheng Street, Longhua District, Shenzhen City, Guangdong Province

Manufacturer : Shenzhen Liyangpeng Technology Co., Ltd

Address : 601, Zone Plant D, 269 Xiangheyuan Industrial, ji ling New Village, ji Tang Community, Fucheng Street, Longhua District, Shenzhen City, Guangdong Province.

Product : Heatsink

Report No: : C211016063001-1

Test method : IEC 62321-3-1:2013& IEC 62321-5:2013&
IEC 62321-4:2013+A1:2017& IEC 62321-7-1:2015
&IEC 62321-7-2:2017 & IEC 62321-6:2015 & IEC 62321-8:2017



Signature: *Will Pan*

Technical director:

Signed for and on behalf of
Euronics Consumer Products Testing Service Co., Ltd

Oct 22, 2021

This certification is part of the full test report(s) and should be read in conjunction with it. The certificate is based on a single evaluation of one sample of above-mentioned products. It does not imply an assessment of the whole production and does not permit the use of the test lab. logo.

Euronics (Dongguan) Consumer Products Testing Service Co., Ltd.
Room 1092, No.12, East Houjie Avenue, Houjie, Dongguan, Guangdong, China

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利洋鹏科技

G48R 规格书

制作版本: A-1

页 次: 19/19

CPST

CERTIFICATE OF CONFORMITY

NO.: C211016062001-2

The following products have been tested by us and found in conformity with the RoHS Directive 2011/65/EU Annex II amending Annex (EU)2015/863 It is possible to use CE marking to demonstrate the conformity with this Directive.

Applicant : Shenzhen Liyangpeng Technology Co., Ltd
Address : 601, Zone Plant D, 269 Xiangheyuan Industrial, ji ling New Village, ji Tang Community, Fucheng Street, Longhua District, Shenzhen City, Guang dong Province.

Manufacturer : Shenzhen Liyangpeng Technology Co., Ltd
Address : 601, Zone Plant D, 269 Xiangheyuan Industrial, ji ling New Village, ji Tang Community, Fucheng Street, Longhua District, Shenzhen City, Guang dong Province.

Product : DC FAN

Model : C211016062001-1
Model/Type reference : 2006,2010,2507,2510,3007,3010,3507,3510,4007,4010,4020,4028,5010,5015,5020,6010,6015,6020,6025,6028,6038,7010,7015,7025,7530,8010,8015, 8020,8025,8038,9025,9225,9015,9038,12015,12032,12038,14025

Report No: : Oct 21 , 2021

Test method : IEC 62321-3-1:2013& IEC 62321-5:2013&
IEC 62321-4:2013+A1:2017& IEC 62321-7-1:2015
&IEC 62321-7-2:2017 & IEC 62321-6:2015 & IEC 62321-8:2017



Signature: *W. J. Pan*
Technical director:
Signed for and on behalf of
Eurones Consumer Products Testing Service Co., Ltd

Oct 21 , 2021

This certification is part of the full test report(s) and should be read in conjunction with it. The certificate is based on a single evaluation of one sample of above-mentioned products. It does not imply an assessment of the whole production and does not permit the use of the test lab. logo.